

(iii) (b), do not allow e.c.f. from (b)(i)2)

- 5 (a) (i)** resistance = V/I C1
= $6.0/(40 \times 10^{-3})$
= 150Ω A1
(no marks for use of gradient)
- (ii)** at 8.0 V, resistance = $8.0/(50 \times 10^{-3}) = 160 \Omega$ C1
change = 10Ω A1 [4]
- (b) (i)** straight line through origin M1
passes through $I = 40 \text{ mA}$, $V = 8.0 \text{ V}$ A1
- (ii)** current in both must be 40 mA C1
e.m.f. = $8.0 + 6.0 = 14.0 \text{ V}$ A1 [4]
- 6 (a) (i)** curve is not smooth, fluctuations, etc B1